

Delta Engine (MKM-DE-001)

Wolfram Oracle Benchmark of FS01, the Risky Annuity and Mark-to-Market

MKM Research Labs

Version 0.1 — 26-October-2026

David K Kelly

CONFIDENTIAL — PROPRIETARY

Legal Notice

PROPRIETARY AND CONFIDENTIAL

This document, including all algorithms, models, methodology, formulas, calculations, and intellectual concepts contained herein, constitutes the exclusive intellectual property and confidential trade secrets of MKM Research Labs ("MKM").

Any usage, reproduction, distribution, modification, or disclosure of this document or any portion thereof, without the express prior written authorisation from MKM Research Labs is strictly prohibited and constitutes an infringement of intellectual property rights.

The document is provided on an "as is" basis. MKM Research Labs makes no representations or warranties of any kind, express or implied, regarding its accuracy, reliability, or suitability for any particular purpose.

All rights reserved. © 2021–2026 MKM Research Labs.

Governed by the laws of the United Kingdom.

Contents

1 Executive Summary **3**

2 Model Purpose and Scope **4**

2.1 What is benchmarked 4

2.2 What is not 4

3 Mathematical Framework **4**

4 Benchmark Method **4**

5 Validation and Backtesting **5**

5.1 The model document's claims 5

5.2 FS01 across the grid 5

6 Model Limitations and Known Weaknesses **6**

7 Document History **6**

1 Executive Summary

This document benchmarks the Delta Engine (MKM-DE-001), which values and risk-manages the PRS book: for every trade it computes the risky annuity, the flood sensitivity FS01 and the mark-to-market. It runs in production, behind the trading routes, the blotter, the XVA service and the historical end-of-day series. The fair spread it differences is PR-001's, benchmarked separately.

The code computes its documented formulae, to the precision it reports. Over 120 cases — annual hazard rates 0.0001–0.5, tenors 1 to 10 years, a flat 3% curve and the book's yield curve, notional GBP 10,000,000 — the risky annuity agrees with a 50-digit reference to 5.8×10^{-16}, and the spread change, FS01 and the returned annuity agree to within half a unit of the rounding the engine applies. Wolfram and the 50-digit reference agree to 4.9×10^{-25}, so the reference is confirmed by a second, independent implementation. The exception is mark-to-market: the engine reuses the annuity *after* rounding it to six decimal places, which moves MTM by up to GBP 3.40 (GBP 0.15 for hazard rates up to 5%). That is immaterial on a GBP 10,000,000 trade, but it is an error, and the fix is to round only for display.

Findings on the model document (Table 2):

1. **FS01 does not scale with the hazard rate.** The document says it scales approximately linearly over 50–500 bp. It barely moves: FS01 at 500 bp is 0.94 times FS01 at 50 bp, not ten times (Table 3). A 1 bp rise in h raises the spread by about 1 bp at any level, so FS01 is set by the risky annuity and the notional.

2. **The yield-curve sensitivity is overstated.** A 100 bp parallel rise in yields lowers FS01 by 0.4–2.4%, not the 3–5% stated.

3. **The hand calculation is wrong.** The 3-year annuity at 200 bp and 4% is 2.724084, not 2.738. The model computes it correctly.

4. **FS01 is blind below the spread floor.** At $h = 0.0001$ the fair spread sits on the 2 bp floor, so the 1 bp bump moves it by 0.0078 bp where the true sensitivity is 1.004 bp: a position on a quiet gauge reports almost no flood risk.

5. **Basis delta is the gauge delta by construction.** The code ignores the basis it is given; the document's approximation is exact only because nothing else is computed.

Two claims hold: the 1 bp bump is within 3.0×10^{-4} bp of the exact gradient (the document says < 0.01 bp), and a 5-year trade carries 4.53 times the FS01 of a 1-year (stated 4.5). A 50 bp move is 0.32% from full revaluation.

Evidence. 7 checks, outcome: 1 fail (Table 1).

- Wolfram: 15.0.0 for Mac OS X ARM (64-bit) (May 26, 2026) via live kernel session (wolframclient).
- Code under test, at commit `7bbb63078a59`:
  `src/models/trading/delta_engine/pricer.py`,
  `src/models/trading/delta_engine/engine.py`.

- Numerics: numpy 1.26.4, scipy 1.15.1, mpmath 1.4.1.
- Generated 2026-10-26 by `python -m tools.wolfram_oracle de001`.

2 Model Purpose and Scope

2.1 What is benchmarked

`compute_risky_annuity`, `compute_gauge_delta`, `compute_basis_delta` and `compute_mark_to_market` in `src/models/trading/delta_engine/pricer.py`, called as `DeltaEngine.enrich_trade` calls them, with the rounding they apply.

2.2 What is not

The fair spread itself (MKM-PR-001's benchmark); the market-state inputs the trader maintains; the risk-grid aggregation, which is a sum. The document's architecture section names `delta_engine.py`, which is now a package. This is a benchmark of the implementation (validation question Q8), not independent validation (Q9).

3 Mathematical Framework

With annual hazard rate h, $\lambda = -\ln(1-h)$, quarterly dates $t_i = i/4$ and discount $D(t) = e^{-r(t)t}$, the document defines

$$A(h,T) = \sum_{i=1}^{4T} \tfrac{1}{4}\, e^{-\lambda t_i} D(t_i), \qquad \mathrm{FS01} = \delta\, \frac{s(h+10^{-4}) - s(h)}{10^4}\, A(h,T)\, N, \qquad \mathrm{MTM} = \delta\, \frac{s - s_0}{10^4}\, A\, N, \tag{1}$$

with s PR-001's fair spread, floored at 2 bp, $\delta = \pm 1$ for a payer or receiver, and N the notional.

Why FS01 is flat in h. On a flat curve PR-001's spread has the closed form $s = 10^4(1-R)\,(e^{\lambda\Delta t} - 1)\, e^{r\Delta t/2}/\Delta t$, so

$$\frac{\mathrm{d}s}{\mathrm{d}h} = 10^4\,(1-R)\,\frac{e^{\lambda\Delta t}\, e^{r\Delta t/2}}{1-h} \approx 10^4. \tag{2}$$

A 1 bp bump in h moves the spread by about 1 bp at any level, and FS01 is about $10^{-4} A\, N$: it follows the annuity, which falls slowly as h rises.

4 Benchmark Method

The equations above are evaluated with `mpmath` at 50 significant digits, from the documents rather than the code; the exact gradient $\mathrm{d}s/\mathrm{d}h$ is a high-precision central difference, checked against the closed form above. Wolfram evaluates the same sums exactly, with the book curve as a piecewise function, differentiates the spread symbolically, and evaluates the bumped spread at the very double the engine computes, $h + 10^{-4}$ in floating point. The engine rounds what it returns, so each rounded output is judged to half a unit in its last place. MTM uses an inception spread of 150 bp.

5 Validation and Backtesting

Table 1: Benchmark checks

Check	Result	Tolerance	Outcome
Risky annuity, unrounded (s3.1)	5.8×10^{-16}	1.0×10^{-12}	pass
Spread change Δs (bp, to 10^{-4}): model vs reference	5.0×10^{-5}	5.0×10^{-5}	pass
FS01 (GBP, to the penny): model vs reference	5.0×10^{-3}	5.0×10^{-3}	pass
Risky annuity as returned (to 10^{-6}): model vs reference	5.0×10^{-7}	5.0×10^{-7}	pass
Mark-to-market (GBP, to the penny): model vs reference	3.4×10^{0}	5.0×10^{-3}	**FAIL**
Wolfram agrees with the reference: annuity, Δs, gradient	4.9×10^{-25}	1.0×10^{-12}	pass
Wolfram agrees with the reference: claims (annuity, FS01)	1.4×10^{-31}	1.0×10^{-12}	pass

5.1 The model document's claims

Table 2: The model document's claims, tested

Claim	Document says	Benchmark finds	Outcome
s6 item 5: 1 bp bump error	$<$ 0.01 bp vs analytical gradient	3.0×10^{-4} bp at most	met
s6 item 1: FS01 against h	"scales approximately linearly", 50–500 bp	FS01(500)/FS01(50) = 0.94, not 10	**not met**
s6 item 2: tenor	5Y $\approx$ 4.5$\times$ 1Y	4.53$\times$	met
s6 item 3: yields +100 bp	FS01 changes 3–5%	FS01 falls 0.4–2.4%	**not met**
s7.2 hand calculation	$A \approx 2.738$	$A = 2.724084$	**not met**
s8 item 1: 50 bp move	FS01$\times\Delta h$ differs from revaluation	by 0.32%	finding
Spread floor (2 bp)	not stated	at $h = 0.0001$: $\Delta s = 0.0078$ bp vs gradient 1.004 bp	finding
s3.5 basis delta	$\approx$ gauge delta	identical in every case: the basis argument is unused	finding

5.2 FS01 across the grid

Table 3: FS01 in GBP from the engine, flat 3% curve, $R = 0$, notional 10,000,000. Down a column it barely moves: FS01 is set by the annuity, not the hazard rate

h	1Y	2Y	3Y	4Y	5Y	10Y
0.0001	8	15	22	29	36	67
0.0005	986	1,941	2,869	3,768	4,640	8,624
0.001	986	1,942	2,868	3,766	4,637	8,609
0.005	988	1,943	2,864	3,754	4,613	8,485
0.01	991	1,944	2,859	3,738	4,583	8,334
0.02	998	1,947	2,849	3,707	4,524	8,043
0.05	1,018	1,956	2,820	3,618	4,353	7,252
0.1	1,053	1,973	2,776	3,477	4,090	6,169
0.2	1,135	2,017	2,701	3,233	3,645	4,673
0.5	1,551	2,303	2,668	2,845	2,931	3,010

6 Model Limitations and Known Weaknesses

1. **Benchmark, not validation.** The engine is shown to compute its documented formulae; whether a parallel 1 bp bump in the annual rate is the right risk measure is a separate question.

2. **Mark-to-market.** Round the annuity for display only, and pass the unrounded value to `compute_mark_to_market`.

3. **The floor.** Where the fair spread is on the 2 bp floor, FS01 should be reported as the analytic sensitivity or flagged, not as near zero.

4. **Documentation.** §6 items 1 and 3 and the §7.2 hand calculation should be corrected to the figures above; §3.5 should say the basis delta *is* the gauge delta; §5.1 should name the package.

7 Document History

Date	Version	Description	Author
26-Oct-2026	0.1	First issue	David K Kelly